DSAA3073: Theories in Data Science

Tutorial 8A: Choose the Benchmark Before the Algorithm

150-minute core tutorial - 15-minute break - 60-minute protected buffer

Wei WANG · HKUST(GZ) · 2026-08-17

Explorative projection

LESSON BACKBONE

When no distribution generates the next outcome, what same-sequence benchmark remains meaningful, and which expert algorithm survives once the perfect-expert assumption is removed?

In earlier statistical arguments, a fixed distribution connected a sample to a population. An online sequence removes that bridge: the next loss may be chosen after a long and adversarial history. We therefore need to settle two matters before choosing an algorithm. What information is available at the instant of commitment? Against which comparator will performance be measured on the realized sequence?

With that protocol fixed, we will test two increasingly tolerant rules. Halving gives a sharp logarithmic mistake bound while a perfect expert remains alive. Deterministic Weighted Majority replaces elimination by a multiplicative penalty and keeps every expert available, but its exact theorem still carries a factor-two-style obstruction.

The three core blocks take **50, 45, and 55 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-50	Repair the round order and build a same-sequence regret ledger	50 min core
50-65	Scheduled break	15 min
65-110	Run Halving, prove its bound, and remove realizability	45 min core
110-165	Derive the exact deterministic Weighted Majority bound and its limitation	55 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

Commit first, compare on the sequence you actually saw

Chapter 7 closes with margin and optimization results, but this Tutorial does not require them at entry. It receives exactly three earlier tools, in order:

No.	Available item	Boundary use
1	Hoeffding's inequality	Recall what an i.i.d. population argument needs; do not apply it to an arbitrary online sequence.
2	AdaBoost's example distribution	Distinguish algorithmic training weights from a distribution that generates future data.
3	AdaBoost's training-error product bound	Reuse a multiplicative proof pattern without calling it an online population guarantee.

IN-CLASS CHECK

Prompt: Reconstruct the three items in order. Which one depends on an i.i.d. sampling bridge, and why is AdaBoost's example distribution not such a bridge?

Hint: Separate a probability distribution generating observations from weights chosen by an algorithm on a fixed training set.

Answer:

> The order is Hoeffding's inequality; AdaBoost's example distribution; and AdaBoost's training-error product bound. Hoeffding supplies a concentration bridge when observations satisfy the stated independence and boundedness assumptions. AdaBoost's distribution is a round-dependent set of weights over already observed training examples. It directs the weak learner; it does not assert that future online losses are i.i.d.

The cards and ledger below come before the formal protocol. First place the four cards in temporal order. Then complete the cumulative columns using only the information that each card makes available.

Card A	Card B	Card C	Card D
observe every expert's current loss	commit to a current prediction or action rule	incur the loss of the committed choice	update from the complete revealed vector
order:	order:	order:	order:

Round	$\ell_t(1)$	$\ell_t(2)$	$\ell_t(3)$	learner $L_t(\mathcal{A})$
1	0	1	1	1
2	1	0	1	1
3	0	1	0	0
4	1	0	0	1
cumulative				

Conclusion card	supported by this ledger?	Reason
the learner has low population risk		
expert 1 is the uniquely best expert		
the learner is one loss behind the best fixed expert on these four rounds		

Table 1: A choose-before-reveal ordering task and a same-sequence loss ledger. Commit to the order and calculations before reading the definitions.

IN-CLASS CHECK

Prompt: Order the cards, total every column, and classify the three conclusion cards. Which current-round quantity must still be hidden when the learner commits?

Hint: The update uses the full vector, but only after the current decision can no longer be changed.

Answer:

> The order is B, A, C, D: commit; reveal all expert losses; incur the committed loss; then update. The expert totals are $2, 2, 2$, and the learner total is 3. Thus the best fixed-expert loss is 2 and the learner is one loss behind it. The three experts tie, so expert 1 is not uniquely best. Nothing in the four-row sequence estimates population risk. The entire current loss vector ℓ_t must be hidden at commitment time.

Definition: Full-information online learning protocol

There are N experts and T rounds. On round t, the learner uses only information from earlier rounds to commit to a current prediction or action rule. The current expert-loss vector is then revealed,

$$\ell_t = (\ell_t(1), \ldots, \ell_t(N)), \tag{1}$$

the learner incurs the scalar round loss $L_t(\mathcal{A})$, and the complete vector is available for the next update. **Full information** means that every coordinate of ℓ_t is observed after commitment, not before it.

The vector above is the registered online-loss notation. Coordinate $\ell_t(i)$ is expert i's loss on round t; $L_t(\mathcal{A})$ is the learner's one scalar loss. In this Tutorial the scalar form is deliberate. A later Tutorial will introduce a distribution notation and a corresponding mixture specialization; no such symbol is needed here.

Definition: Non-anticipating adversary

The current loss vector may depend on the past and on the learner's known rule. If the learner will later use a private current random sample, however, the loss vector is fixed without observing that sample. Access to the public rule is allowed; access to the private current draw before fixing ℓ_t is not. The private-draw exclusion is an explicit course information assumption. The source-supported round order is commitment before the current cost is revealed.

IN-CLASS CHECK

Prompt: Classify each adversary: (i) chooses ℓ_t from the entire past; (ii) knows the learner's public current rule but not its private current sample; (iii) waits to see that private sample and only then fixes ℓ_t.

Hint: The boundary concerns current private information, not dependence on the past.

Answer:

> Cases (i) and (ii) satisfy the declared non-anticipating information condition. Case (iii) does not. The first two may still be highly adaptive: they can exploit the complete history and the public algorithm. What they cannot exploit is a private current sample that has not been revealed.

The protocol supplies losses but not yet a performance target. Population risk would require a distributional bridge that the online model deliberately does not assume. The four-round ledger suggests a comparator that needs no such bridge: evaluate learner and experts on the same realized sequence.

Definition: External regret

The **external regret** of learner $\mathcal{A}$ after T rounds is

$$R_T = \sum_{t=1}^{T} L_t(\mathcal{A}) - \min_{i \in \{1,\dots,N\}} \sum_{t=1}^{T} \ell_t(i). \tag{2}$$

The minimum chooses one fixed expert after the sequence is visible, then evaluates that expert on every round of the same sequence. It is not a round-by-round switching oracle, and R_T is not population risk.

Worked example: One sequence, one fixed comparator

In the four-round ledger, each fixed expert loses 2 while the learner loses 3, so $R_4 = 3 - 2 = 1$ and $\frac{R_4}{4} = 0.25$. Selecting the lower-loss expert anew on each row would give loss 0, but that switching benchmark is absent from the definition. The tie among all three fixed experts also shows why a regret value does not identify one distinguished expert.

Definition: Sublinear regret

Regret is **sublinear** when $R_T = o(T)$, equivalently $\frac{R_T}{T} \to 0$ as $T \to \infty$. The conclusion is only that the learner's average loss approaches the average loss of the best fixed expert on the realized sequence. It does not say that either average loss is small, that a unique expert has been found, or that a new i.i.d. observation will be predicted well.

IN-CLASS CHECK

Prompt: For each regret sequence, decide whether the guarantee is sublinear and state the average excess loss: $R_T = \sqrt{T}$, $R_T = 0.1T$, and $R_T = \log T$.

Hint: Divide each expression by T; a small fixed coefficient is not the same as a vanishing coefficient.

Answer:

For $\sqrt{T}$, the average excess is $\frac{1}{\sqrt{T}} \to 0$, so the guarantee is sublinear. For $0.1T$, the average excess remains 0.1, so it is not sublinear. For $\log T$, the average excess is $\frac{\log T}{T} \to 0$, so it is sublinear. None of the three statements alone controls absolute loss or population risk.

We now have an honest benchmark. The first algorithm earns a logarithmic guarantee only by making a much stronger claim about the experts themselves.

Let consistency shrink the candidate set

A **perfect expert** is an expert whose loss is zero on every round of the complete binary prediction sequence. This is a realizability assumption, not the automatic existence of a hindsight minimizer. Every finite sequence has a best expert; it need not have a perfect one.

Before naming the algorithm, complete the paired ledgers. A row marked "learner mistake" retains only experts that predicted the revealed binary label. In the left ledger, one unnamed expert is promised to be perfect. In the right ledger, the two experts always disagree and no such promise is made.

Round	**active before**	**votes 0 / 1**	**majority prediction**	y_t	**active after**
1	8	$\frac{5}{3}$	0	1	
2		$\frac{1}{2}$	1	0	
proof card	$\lvert S_t\rvert$	mistaken majority			$\lvert S_{t+1}\rvert \le$

Round	h_1	h_2	**tie rule / prediction**	y_t	**active after**
1	0	1	tie $\to$ 0	1	
2	inactive	1	1	0	
failed proof card					which lower bound disappears?

Table 2: A realizable active-set shrinkage trace and a nonrealizable empty-set trace. Locate both the halving inequality and the surviving-expert lower bound.

IN-CLASS CHECK

Prompt: Complete both ledgers. On the left, which inequality follows from a learner mistake? On the right, exactly when does the active set become empty?

Hint: A wrong majority contains at least half of the active experts; the perfect expert is the witness that survives every update.

Answer:

On the left, round 1 retains the three experts voting 1; round 2 retains the one expert voting 0. Whenever the majority prediction is wrong, all experts in that mistaken majority are removed, so $|S_{t+1}| \le |S_t|\frac{1}{2}$. The promised perfect expert agrees with every revealed label and remains in the active set. On the right, round 1 removes h_1 and leaves h_2; round 2 removes h_2, leaving the empty set. The missing proof line is the lower bound $|S_t| \ge 1$.

Definition: Halving algorithm

Begin with all N binary experts active. On each round, predict by majority vote among the active experts, using a fixed tie rule. After the true label is revealed, retain exactly the active experts that predicted that label if the learner made a mistake; after a correct learner prediction, leave the active set unchanged. Under realizability, a perfect expert remains active throughout.

Result: Halving mistake bound under realizability

Suppose $N \geq 1$ binary experts are available and at least one expert is perfect on the complete sequence. If Halving makes m_T mistakes, then

$$1 \leq |S_{T+1}| \leq \frac{N}{2^{m_T}}, \tag{3}$$

and therefore

$$m_T \leq \log_2 N. \tag{4}$$

This is a mistake bound conditional on a surviving perfect expert. Because that expert has cumulative loss zero, external regret happens to equal m_T in this realizable zero-one setting; the theorem is not a general nonrealizable regret guarantee.

The proof uses two logically separate lines. The majority mistake supplies the upper shrinkage route. Realizability supplies the nonempty lower route. Remove the latter and repeated elimination can leave no prediction rule at all.

IN-CLASS CHECK

Prompt: With $N = 32$ and a perfect expert, what is the largest mistake count allowed by the theorem? If the best expert instead makes one mistake, may you simply add one to this answer?

Hint: The theorem's proof keeps one expert alive forever; one comparator mistake destroys that witness.

Answer:

> The bound gives $m_T \leq \log_2 32 = 5$. If the best expert makes one mistake, the realizability assumption is false and that expert can be eliminated on its mistaken round. The proof no longer supplies a nonempty active set, so adding one to the bound is unjustified.

IN-CLASS CHECK

Prompt: In the two-expert empty-set trace, compute the best fixed expert's loss after the two rounds and explain why "a best expert exists" does not repair Halving.

Hint: Compare the minimum cumulative loss with the stronger requirement of zero loss on every prefix.

Answer:

> Expert h_1 is wrong only on round 1 and h_2 only on round 2, so the best fixed loss is 1. A best expert therefore exists, but neither expert is perfect. Halving removes h_1 on the first label and h_2 on the second; hindsight optimality does not supply an expert that survives every prefix.

The empty set is a brittle failure. A natural repair is to let mistaken experts lose influence without deleting them. That repair changes the proof and the quality of the guarantee.

IN-CLASS CHECK

Prompt: Before choosing the repair, separate three desired claims: (i) the learner always has experts left to consult; (ii) the mistake bound compares with an imperfect best expert; (iii) the resulting external regret is sublinear. Would keeping every expert at positive weight prove all three?

Hint: A state invariant, a comparator theorem, and an additive regret rate are three different obligations.

Answer:

> Positive finite-round weights would establish only the first claim: the learner never reaches an empty weighted state. A new proof is still needed to compare the learner's mistakes with an imperfect best expert. Even such a comparison might be multiplicative rather than $m_T^* + o(T)$, so sublinear external regret would remain a separate question.

Penalize mistakes without erasing an expert

Use the same two always-disagreeing experts, a fixed tie rule that predicts 0, and penalty $\beta = \frac{1}{2}$. Complete the table before reading the update rule. Every mistaken expert is multiplied by β after the revealed label.

Round	$w_t(1)$	$w_t(2)$	weighted prediction	y_t	learner loss	weights after update
1	1	1	0	1	1	
2				0		
3				1		

Proof route	Quantity to bound	Blank relation
learner-mistake route	total expert weight W_t	$W_{t+1} \le$
fixed-comparator route	one expert's surviving weight	$W_{T+1} \ge$
adaptive lower example	two experts that always disagree	$m_T - m_T^* \ge$

Table 3: A soft-penalty trace and the three relations that will determine both the theorem and its deterministic limitation.

IN-CLASS CHECK

Prompt: Complete all three rounds. What failure of Halving has disappeared, and what learner behavior in this trace has not disappeared?

Hint: A multiplicative penalty in $(0, 1)$ keeps weights positive, but a deterministic visible vote can still be made wrong.

Answer:

After round 1 the weights are $(\frac{1}{2}, 1)$. Round 2 therefore predicts 1, is wrong on label 0, and updates to $(\frac{1}{2}, \frac{1}{2})$. Round 3 ties, predicts 0, is wrong on label 1, and updates to $(\frac{1}{4}, \frac{1}{2})$. No expert is deleted, so the empty-set failure is gone. The learner is nevertheless wrong on every displayed round because each deterministic current prediction is visible before the adversarial label is selected.

Definition: Deterministic Weighted Majority

Fix $\beta \in (0, 1)$ and initialize $w_1(i) = 1$ for every binary expert. On round t, predict by the current weighted majority, with a fixed tie rule. After the label is revealed, keep a correct expert's weight unchanged and update each mistaken expert by

$$w_{t+1}(i) = \beta w_t(i). \tag{5}$$

This soft penalty leaves every finite-round weight positive. It does not by itself imply additive sublinear external regret.

Let $W_t = \sum_{i=1}^{N} w_t(i)$ be the total expert weight before round t. Here it is a proof bookkeeping device. On a learner mistake, at least half of W_t lies on experts that made the same wrong prediction. Multiplying that part by β gives

$$W_{t+1} = W_t - (1-\beta)W_t^{\text{wrong}} \le \left(1 - \frac{1-\beta}{2}\right)W_t = \frac{1+\beta}{2}W_t. \tag{6}$$

On other rounds total weight cannot increase. Hence, after m_T learner mistakes,

$$W_{T+1} \le N\left(\frac{1+\beta}{2}\right)^{m_T}. \tag{7}$$

If the best fixed expert makes m_T^* mistakes, its final weight is $\beta^{m_T^*}$. Since that one nonnegative weight is part of the total,

$$\beta^{m_T^*} \le W_{T+1} \le N\left(\frac{1+\beta}{2}\right)^{m_T}. \tag{8}$$

Taking logarithms and keeping their signs straight produces the exact result.

Result: Mohri's exact deterministic Weighted Majority bound

For $N \ge 1$, $T \ge 1$, $\beta \in (0,1)$, binary expert predictions, and zero-one loss, let m_T be the deterministic Weighted Majority mistake count and m_T^* the mistake count of the best single fixed expert in hindsight. Then

$$m_T \le \frac{\ln N + m_T^* \ln\left(\frac{1}{\beta}\right)}{\ln\left(\frac{2}{1+\beta}\right)}. \tag{9}$$

The theorem applies without a perfect expert. In the realizable case $m_T^* = 0$, it reduces to $m_T \le \ln \frac{N}{\ln\left(\frac{2}{1+\beta}\right)}$; for positive comparator loss, the exact beta-dependent multiplicative term remains.

IN-CLASS CHECK

Prompt: Reproduce both weight routes and solve for m_T. Why does dividing by $\ln\left(\frac{1+\beta}{2}\right)$ directly invite a sign error?

Hint: That logarithm is negative; rewrite the final denominator as the positive quantity $\ln\left(\frac{2}{1+\beta}\right)$.

Answer:

The upper route is $W_{T+1} \le N\left(\frac{1+\beta}{2}\right)^{m_T}$ and the comparator route is $W_{T+1} \ge \beta^{m_T^*}$. Thus $\beta^{m_T^*} \le N\left(\frac{1+\beta}{2}\right)^{m_T}$. After taking logarithms and moving the negative terms, we obtain $m_T \ln\left(\frac{2}{1+\beta}\right) \le \ln N + m_T^* \ln\left(\frac{1}{\beta}\right)$, which yields the displayed bound. Dividing by the negative logarithm $\ln\left(\frac{1+\beta}{2}\right)$ without reversing the inequality would be incorrect.

Worked example: Evaluate the exact formula, not a slogan

Let $N = 8$, $\beta = \frac{1}{2}$, and $m_T^* = 2$. Then

$$m_T \le \frac{\ln 8 + 2\ln 2}{\ln\left(\frac{4}{3}\right)} = \frac{\ln 32}{\ln\left(\frac{4}{3}\right)} \approx 12.05. \tag{10}$$

Since m_T is an integer, the theorem certifies $m_T \le 12$. If instead $m_T^* = 0$, it gives $m_T \le \ln \frac{8}{\ln\left(\frac{4}{3}\right)} \approx 7.23$, hence at most 7 mistakes. This realizable specialization is valid but looser than Halving's $\log_2 8 = 3$ guarantee because $\beta > 0$ preserves mistaken experts.

IN-CLASS CHECK

Prompt: For $N = 8$ and $\beta = \frac{1}{2}$, calculate the realizable specialization and compare it with Halving. Which algorithm has the stronger bound, and which remains defined after every expert has made a mistake?

Hint: Strength under realizability and robustness without realizability are different criteria.

Answer:

Deterministic Weighted Majority gives $m_T \le \ln \frac{8}{\ln\left(\frac{4}{3}\right)} \approx 7.23$, so its integer mistake count is at most 7. Halving gives at most 3 mistakes and is stronger while a perfect expert exists. Weighted Majority keeps every weight positive for finite time and therefore remains defined when all experts have made mistakes; Halving can reach the empty set.

Soft penalties repair the crash, but they do not conceal a deterministic current prediction. Let expert 1 always predict 0 and expert 2 always predict 1. After the learner commits, choose the opposite binary label. The learner makes $m_T = T$ mistakes. Across the same T labels, at least one of the two constant experts makes at most $\frac{T}{2}$ mistakes, so

$$R_T = m_T - m_T^* \ge \frac{T}{2}. \tag{11}$$

This construction respects choose before reveal. It needs no access to a private current sample because the learner is deterministic. It also does not contradict the exact theorem: the beta-dependent coefficient multiplying the best expert's mistakes has a factor-two-style lower limit.

IN-CLASS CHECK

Prompt: Explain why the two-expert sequence proves linear regret for every deterministic zero-one learner yet does not refute the exact Weighted Majority theorem.

Hint: Separate an additive $m_T^* + o(T)$ target from a multiplicative comparison with coefficient at least two.

Answer:

The adversary makes the deterministic learner wrong on every round, while one of the two constant experts is wrong on at most half the rounds. Hence external regret is at least $\frac{T}{2}$ and cannot be sublinear. Mohri's theorem does not promise $m_T \le m_T^* + o(T)$; its exact coefficient on m_T^* is beta-dependent and never removes the factor-two-style limitation. Linear regret can therefore coexist with the theorem.

The remaining question is now precise. Can the learner hide the current realized expert choice while preserving the choose-before-reveal information boundary, so that the comparison becomes additive in expectation? That work belongs to Tutorial 8B.

Standard language to carry forward

Term or notation	Meaning here	Representative form or condition
full-information online protocol	Commit before the current loss vector is revealed; observe the complete vector afterward.	choose $\to$ reveal ℓ_t $\to$ incur $\to$ update
non-anticipating adversary	May use the past and public rule, but not a private current sample before fixing the loss.	private-current-draw exclusion is an explicit information assumption
$\ell_t = (\ell_t(1), \ldots, \ell_t(N))$	Registered vector of expert losses on round t.	full vector is post-commitment feedback
external regret	Learner loss minus the best one fixed expert on the same realized sequence.	$R_T = \sum_t L_t(\mathcal{A}) - \min_i \sum_t \ell_t(i)$
sublinear regret	Average excess loss against that comparator vanishes.	$R_T = o(T)$ or $\frac{R_T}{T} \to 0$
Halving	Majority elimination with a logarithmic mistake bound only while a perfect expert survives.	$m_T \le \log_2 N$ under realizability
deterministic Weighted Majority	Weighted vote with mistaken-expert penalty $\beta \in (0,1)$.	$w_{t+1}(i) = \beta w_t(i)$ for a mistaken expert
exact deterministic WM bound	A beta-dependent multiplicative mistake comparison, not additive sublinear regret.	$m_T \le \frac{\ln N + m_T^* \ln\left(\frac{1}{\beta}\right)}{\ln\left(\frac{2}{1+\beta}\right)}$

Reconstruct the exact Tutorial 8B entry

Tutorial 8B receives eleven items. The first three pass through from the Chapter entry; the remaining eight were established here. Their order is part of the boundary.

No.	Available item	What Tutorial 8B may assume
1	Hoeffding's inequality	Recognize an i.i.d. concentration tool and keep it separate from arbitrary-sequence regret.
2	AdaBoost's example distribution	Recognize an algorithmic distribution over fixed training examples.
3	AdaBoost's training-error product bound	Reuse the two-route multiplicative proof pattern without identifying the algorithms.
4	full-information online protocol	Execute commit, reveal the full loss vector, incur scalar learner loss, and update.
5	non-anticipating adversary	State the private-current-sample information boundary.
6	external regret	Compute against one best fixed expert on the same sequence.
7	sublinear regret	Interpret only the vanishing average excess loss.
8	Halving	Apply its logarithmic mistake proof only under a surviving perfect expert.
9	deterministic Weighted Majority	Run the beta update and interpret the exact factor-two-style limitation.
10	$\ell_t = (\ell_t(1), ..., \ell_t(N))$	Decode the registered online loss vector.
11	$R_T = \sum_t L_t(\mathcal{A}) - \min_i \sum_t \ell_t(i)$	Use the registered scalar external-regret form before any later specialization.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct all eleven items in order. Then answer: which item blocks reveal-before-choose, which item blocks a population-risk interpretation, and which two algorithms separate realizability from robustness?

Hint: The protocol controls time, regret controls the comparator, and the two deterministic algorithms make different assumptions.

Answer:

The order is: Hoeffding's inequality; AdaBoost's example distribution; AdaBoost's training-error product bound; full-information online protocol; non-anticipating adversary; external regret; sublinear regret; Halving; deterministic Weighted Majority; the registered loss vector $\ell_t = (\ell_t(1), \dots, \ell_t(N))$; and the registered scalar external-regret form $R_T = \sum_t L_t(\mathcal{A}) - \min_i \sum_t \ell_t(i)$. The protocol enforces choose before reveal. External regret keeps the claim on the realized sequence rather than a population. Halving supplies the sharper realizable bound; deterministic Weighted Majority remains defined without a perfect expert but retains the factor-two-style limitation.

The core Tutorial ends here. The optional Chapter 8 application, fully adaptive-adversary, and exponential-update comparisons are deliberately not inserted into Tutorial 8A; the approved paired plan places one answered, zero-minute appendix only after Tutorial 8B's complete core and Chapter 9 handoff.

Source note

The full-information protocol, scalar external-regret benchmark, Halving algorithm, realizable mistake proof, deterministic Weighted Majority update, exact beta-dependent bound, and two-expert deterministic obstruction follow Mohri, Rostamizadeh, and Talwalkar, **Foundations of Machine Learning**, second edition, Sections 8.1-8.2.2 and the opening of Section 8.2.3 (printed pp. 178-183; PDF pp. 195-200). The distribution-before-cost order and expected mixture-cost framework consulted for the later randomized boundary are from Arora, Hazan, and Kale, Section 2 (journal pp. 126-128; PDF pp. 6-8). Excluding access to a learner's private current sample is stated here as an explicit course information assumption, not as a quotation from Hazan Section 1.1. Hazan Section 1.1 (printed pp. 2-3; PDF pp. 24-25) supports the choose-before-reveal online framing and the sublinear-regret interpretation. The historical Week 8 Learning Sheet controls topic scope only; claims and proofs above are reconstructed under the original sources.

DSAA3073: Theories in Data Science

Tutorial 8B: From a Hidden Draw to a Regret Proof

150-minute core tutorial - 15-minute break - 60-minute protected buffer

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Explorative projection

LESSON BACKBONE

How does a distribution chosen before the current losses defeat a deterministic attack, and which exact weight identity turns that idea into a continuous-loss regret theorem?

Tutorial 8A left us with a precise obstruction. A deterministic binary prediction can be observed and opposed on every round. Randomization helps only if we say what remains hidden, what loss is averaged, and where the expectation enters the guarantee. We will settle those questions first.

We then replace the binary mistake penalty by the course-primary linear multiplicative update. One total-weight identity gives an upper route from the learner's mixture cost; one fixed expert gives a lower route. Their logarithms produce the comparator theorem. The details matter: the theorem uses a positive capped learning rate, and the linear factor is not an interchangeable spelling of an exponential update, AdaBoost, or a gradient step.

The three core blocks take **65, 65, and 20 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge. Optional comparisons appear only after the complete core and consume zero class minutes.

Clock	Mathematical work	Protected time
0-65	Hide the current sample and derive Randomized Weighted Majority	65 min core
65-80	Scheduled break	15 min
80-145	Generalize to bounded costs and prove the linear MWU theorem	65 min core
145-165	Select the valid guarantee and reconstruct the Chapter 9 entry	20 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

Hide the current draw, not the public distribution

Tutorial 8B receives exactly eleven items. Reconstruct them before the new randomized rule is shown.

No.	Available item	Use at today's boundary
1	Hoeffding's inequality	Recognize the assumptions of an i.i.d. concentration argument; do not apply it to an arbitrary online sequence.
2	AdaBoost's example distribution	Recognize algorithmic weights on fixed training examples.
3	AdaBoost's training-error product bound	Reuse a potential-proof pattern without identifying two algorithms.
4	full-information online protocol	Commit before reveal and observe the complete current loss vector afterward.
5	non-anticipating adversary	Exclude access to the learner's private current sample before the loss is fixed.
6	external regret	Compare with one best fixed expert on the same realized sequence.
7	sublinear regret	Interpret a vanishing average excess loss only.
8	Halving	Use its logarithmic mistake proof only while a perfect expert survives.
9	deterministic Weighted Majority	Use its soft penalty and exact beta-dependent bound without claiming additive regret.
10	$\ell_t = (\ell_t(1), \ldots, \ell_t(N))$	Decode the registered full loss vector.
11	$R_T = \sum_t L_t(\mathcal{A}) - \min_i \sum_t \ell_t(i)$	Start from the registered scalar regret form.

IN-CLASS CHECK

Prompt: Reconstruct the eleven items in order. Which item controls the current information boundary, and which formula still contains no online distribution?

Hint: The fifth item governs private current information; the final formula uses the learner's scalar round loss.

Answer:

> The order is Hoeffding; AdaBoost's example distribution; AdaBoost's training-error product bound; the full-information protocol; the non-anticipating adversary; external regret; sublinear regret; Halving; deterministic Weighted Majority; the registered loss vector; and the registered scalar external-regret form. The non-anticipating condition protects the current private sample. The last formula contains only the learner's scalar loss; the online distribution notation is introduced below before it is used.

Return to two experts that always disagree. Their current weights are 3 and 1. A deterministic weighted vote announces expert 1's prediction. Instead, complete the probability blanks and put the four round cards in a legal order.

Expert	current weight	sampling probability	current zero-one loss
1	3		$\ell_t(1)$
2	1		$\ell_t(2)$
total	4		

Card A	Card B	Card C	Card D
normalize the public weights	sample one expert privately	fix and reveal the current loss vector without seeing the sample	incur the sampled loss and update every revealed coordinate
order:	order:	order:	order:

Claim card	Status	Reason
the sampled loss is always the weighted average		
the conditional expected sampled loss is the weighted average		

Table 4: A hidden-sample timeline and expectation ledger. The student acts on weights and information order before the distribution notation is defined.

IN-CLASS CHECK

Prompt: Complete the probabilities and card order. If the revealed loss vector is $(1, 0)$, what are the two possible sampled losses, and which one equals their weighted average?

Hint: A random realization is one coordinate; an expectation averages both coordinates using the pre-loss probabilities.

Answer:

The probabilities are $\frac{3}{4}$ and $\frac{1}{4}$, and the order is A, B, C, D. With loss vector $(1, 0)$, the realized sampled loss is 1 if expert 1 was sampled and 0 if expert 2 was sampled. Neither outcome must equal the weighted average $\frac{3}{4}$. The weighted average is the conditional expectation $\left(\frac{3}{4}\right)1 + \left(\frac{1}{4}\right)0 = \frac{3}{4}$.

Definition: Randomized Weighted Majority and the online distribution

Let the current expert weights be positive and define the registered online distribution

$$p_t(i) = \frac{w_t(i)}{\sum_{j=1}^{N} w_t(j)}, \quad i = 1, \ldots, N. \tag{12}$$

This distribution is formed before the current loss vector is fixed or revealed. The learner may sample I_t privately from p_t and incur $\ell_t(I_t)$, or equivalently account for the mixture loss

$$L_t = \sum_{i=1}^{N} p_t(i)\ell_t(i). \tag{13}$$

After the full zero-one loss vector is revealed, a mistaken expert is multiplied by β and a correct expert is unchanged. The current private sample is not revealed to the adversary before ℓ_t is fixed.

The symbol p_t is now available. It is an online distribution over experts, not AdaBoost's D_t over fixed training examples. The two objects are both normalized weights, but they live on different sets, control different rounds, and support different theorems.

Let $\mathcal{F}_{t-1}$ denote all public history before the current draw. Because ℓ_t is fixed without observing I_t,

$$E[\ell_t(I_t) \mid \mathcal{F}_{t-1}, p_t, \ell_t] = \sum_{i=1}^{N} p_t(i)\ell_t(i) = L_t. \tag{14}$$

This is a conditional-expectation equality. It does not say $\ell_t(I_t) = L_t$ on each realization. Summing and taking expectation again therefore turns a deterministic mixture-loss theorem into an expected sampled- loss theorem under the declared information pattern.

IN-CLASS CHECK

Prompt: Suppose $p_t = (0.2, 0.5, 0.3)$ and $\ell_t = (1, 0, 1)$. Compute L_t and list every possible sampled loss. Then state exactly where expectation is needed.

Hint: The mixture calculation is deterministic once the distribution and loss vector are fixed; only the sampled coordinate is random.

Answer:

The mixture loss is $L_t = 0.2 + 0 + 0.3 = 0.5$. The sampled loss can be 1, 0, or 1 depending on the sampled expert. Conditional on the pre-loss distribution and the fixed vector, its expectation is 0.5. Expectation is needed when translating from the sampled path to the mixture; the arithmetic value of the mixture itself is not random after conditioning.

The exact beta-dependent theorem follows from the same potential W_t used in Tutorial 8A, but the learner-loss route changes. With binary losses,

$$W_{t+1} = \sum_{i} w_t(i)\beta^{\ell_t(i)} = W_t(1 - (1-\beta)L_t). \tag{15}$$

Thus $W_{T+1} = N\prod_{t=1}^{T}(1-(1-\beta)L_t)$. If i^* is the best fixed expert and $L_T^* = \sum_t \ell_t(i^*)$, then its surviving weight supplies $W_{T+1} \ge \beta^{L_T^*}$. Taking logarithms, using $\ln(1-x) \le -x$, and using $-\ln\beta \le (1-\beta)(2-\beta)$ for $\beta \in \left[\frac{1}{2}, 1\right)$ gives the result.

Result: Randomized Weighted Majority guarantee

Let $N \ge 1$, $T \ge 1$, binary losses $\ell_t(i) \in \{0,1\}$, and $\beta \in \left[\frac{1}{2}, 1\right)$. For the cumulative mixture loss $L_T = \sum_t \sum_i p_t(i)\ell_t(i)$ and best fixed-expert loss $L_T^* = \min_i \sum_t \ell_t(i)$,

$$L_T \le \frac{\ln N}{1-\beta} + (2-\beta)L_T^*. \tag{16}$$

Equivalently,

$$L_T - L_T^* \le \frac{\ln N}{1-\beta} + (1-\beta)L_T^*. \tag{17}$$

For $N \ge 2$ and known horizon, choosing $\beta = \max\left\{\frac{1}{2}, 1 - \sqrt{\ln\frac{N}{T}}\right\}$, together with the trivial bound $L_T - L_T^* \le T$ when the cap is active, yields

$$L_T - L_T^* \le 2\sqrt{T\ln N}. \tag{18}$$

Under the non-anticipating private-draw model, the same statements hold for cumulative sampled loss in expectation. They are not pathwise promises for every sequence of coin outcomes.

IN-CLASS CHECK

Prompt: Starting from $W_{T+1} \ge \beta^{L_T^*}$ and the product upper route, recover the first displayed theorem. Which sentence prevents you from claiming it for every sampled path?

Hint: Take logarithms before substituting the inequality that controls $-\ln\beta$; then distinguish mixture loss from sampled loss.

Answer:

The two routes give $L_T^*\ln\beta \le \ln N + \sum_t \ln(1-(1-\beta)L_t) \le \ln N - (1-\beta)L_T$. Rearranging gives $L_T \le \ln\frac{N}{1-\beta} - \ln(\beta)\frac{L_T^*}{1-\beta}$. For $\beta \in \left[\frac{1}{2}, 1\right)$, $-\ln\frac{\beta}{1-\beta} \le 2-\beta$, giving the theorem. The final translation to sampled loss uses conditional expectation; a realized sample can lie above or below its mixture loss on any round, so no every-path conclusion follows.

Worked example: Tune beta without losing the expectation label

Let $N = 16$ and $T = 400$. Then $\sqrt{\frac{\ln(16)}{400}} \approx 0.0833$, so $\beta \approx 0.9167$. The additive expected-regret guarantee is at most $2\sqrt{400\ln 16} \approx 66.60$. This number bounds cumulative mixture regret, and therefore expected sampled regret under the information condition. It does not assert that every random path has regret at most 66.60.

Randomized weighting solves the deterministic binary obstruction. The proof still treats losses as either zero or one. The next question is whether the same distribution can absorb every loss between those endpoints while keeping an exact comparator theorem.

Let every loss set its own penalty

Before naming the update, take penalty scale $\eta = \frac{1}{4}$ and current weights $(2, 1, 1)$. The three experts incur costs $\left(0, \frac{1}{2}, 1\right)$. Complete the first table with a multiplicative factor that is 1 at cost zero and $1 - \eta$ at cost one, and varies linearly between them. Then use the completed row to fill the three proof lanes.

Expert	$w_t(i)$	cost	linear factor	new weight
1	2	0		
2	1	$\frac{1}{2}$		
3	1	1		

Lane	Quantity	Relation to recover
learner mixture	$\sum_i p_t(i)\ell_t(i)$	
total weight	$\Phi_t = \sum_i w_t(i)$	$\Phi_{t+1} =$
fixed comparator	$w_{T+1}(i)$	$\ln(w_{T+1}(i)) \geq$

Domain card	Legal?	Reason
$\eta = 0$		
$\eta = \frac{3}{4}$		
$0 < \eta \leq \frac{1}{2}$		

Table 5: A pre-definition continuous-penalty trace, followed by the learner, potential, comparator, and theorem-domain lanes.

IN-CLASS CHECK

Prompt: Complete the three expert rows. Normalize the old weights, compute the mixture cost, and verify the exact total-weight identity numerically.

Hint: Use factors $1 - \eta\ell_t(i)$ and normalize before the update.

Answer:

The factors are 1, $\frac{7}{8}$, and $\frac{3}{4}$, so the new weights are $\left(2, \frac{7}{8}, \frac{3}{4}\right)$. The old total is $\Phi_t = 4$ and $p_t = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$. The mixture cost is $0 + \left(\frac{1}{4}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{4}\right)1 = \frac{3}{8}$. The new total is $\frac{29}{8} = 3.625$, which equals $4\left(1 - \left(\frac{1}{4}\right)\left(\frac{3}{8}\right)\right) = 4\left(\frac{29}{32}\right) = \frac{29}{8}$.

Definition: Linear Multiplicative Weights Update

Let n be a positive integer and initialize $w_1(i) = 1$. On round t, normalize the current weights to the already defined distribution $p_t(i) = \frac{w_t(i)}{\Phi_t}$, where $\Phi_t = \sum_i w_t(i)$. Commit using p_t, observe the complete cost vector $\ell_t \in [-1, 1]^n$, and update

$$w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i)). \tag{19}$$

The exported theorem uses $0 < \eta \le \frac{1}{2}$. Hence every factor lies in $[1 - \eta, 1 + \eta] \subset (0, \infty)$, so finite-round weights and Φ_t remain positive. This linear factor is the course-primary update.

The restriction $\eta > 0$ is not cosmetic: the regret theorem divides by η. The cap $\eta \le \frac{1}{2}$ licenses the logarithmic inequalities below. A cost may be negative in the signed theorem, in which case its factor exceeds one; the loss-only corollary later specializes to $[0, 1]$.

Definition: Weight-potential analysis

Track the total weight $\Phi_t = \sum_i w_t(i)$ along two routes. The learner mixture gives an upper bound on Φ_{T+1}. Any one fixed comparator's surviving weight gives a lower bound on the same quantity. Combining those bounds after taking logarithms produces a comparator guarantee.

The learner route begins with an equality, not an approximation:

$$\Phi_{t+1} = \sum_i w_t(i)(1 - \eta\ell_t(i)) = \Phi_t \left(1 - \eta \sum_i p_t(i)\ell_t(i)\right). \tag{20}$$

Write $L_t = \sum_i p_t(i)\ell_t(i)$. Since $1 - x \le e^{-x}$,

$$\Phi_{T+1} \le n \exp\left(-\eta \sum_{t=1}^{T} L_t\right). \tag{21}$$

For a fixed comparator i, the theorem domain gives the elementary inequality

$$\ln(1 - \eta x) \ge -\eta x - \eta^2|x|, \quad x \in [-1, 1], \quad 0 < \eta \le \frac{1}{2}. \tag{22}$$

Applying it to every factor in that comparator's weight gives the lower route

$$\ln(\Phi_{T+1}) \ge \ln(w_{T+1}(i)) \ge -\eta \sum_{t=1}^{T} \ell_t(i) - \eta^2 \sum_{t=1}^{T} |\ell_t(i)|. \tag{23}$$

The two routes now meet. Taking the logarithm of the learner upper route and placing the comparator lower route beneath it yields

$$-\eta \sum_t \ell_t(i) - \eta^2 \sum_t |\ell_t(i)| \le \ln(\Phi_{T+1}) \le \ln n - \eta \sum_t L_t. \tag{24}$$

Rearrangement gives the exact comparator form.

Result: Linear MWU comparator and regret bounds

Let n be a positive integer, $T \geq 1$, $\ell_t(i) \in [-1, 1]$, and $0 < \eta \leq \frac{1}{2}$. The linear update $w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i))$ guarantees, for every fixed expert i,

$$\sum_{t=1}^{T} p_t \cdot \ell_t \leq \sum_{t=1}^{T} \ell_t(i) + \eta \sum_{t=1}^{T} |\ell_t(i)| + \frac{\ln n}{\eta}. \tag{25}$$

Here $p_t \cdot \ell_t = \sum_i p_t(i)\ell_t(i)$. For losses in $[0, 1]$, comparison with the best fixed expert gives the coarser external-regret bound

$$R_T \leq \eta T + \frac{\ln n}{\eta}. \tag{26}$$

If an expert is sampled privately from p_t, the learner side is expected cumulative loss under the same non-anticipating information condition used above.

IN-CLASS CHECK

Prompt: Reproduce the exact one-step identity, the accumulated learner upper route, and the fixed-comparator lower route. Which term would be lost by prematurely replacing every $|\ell_t(i)|$ with 1?

Hint: Keep the comparator-dependent magnitude term until the signed-cost theorem is complete.

Answer:

> The identity is $\Phi_{t+1} = \Phi_t(1 - \eta p_t \cdot \ell_t)$, so $\Phi_{T+1} \leq n \exp\left(-\eta \sum_t p_t \cdot \ell_t\right)$. For any fixed i, $\ln(\Phi_{T+1}) \geq -\eta \sum_t \ell_t(i) - \eta^2 \sum_t |\ell_t(i)|$. Combining and dividing by positive η yields the theorem. Replacing magnitudes by 1 too early discards the sharper $\eta \sum_t |\ell_t(i)|$ comparator term and leaves only the coarser ηT corollary.

Worked example: A signed comparator can gain weight

Let $\eta = \frac{1}{4}$ and let one fixed expert's three costs be $\left(-1, \frac{1}{2}, 0\right)$. Its exact multiplicative factor over the three rounds is $\left(1 + \frac{1}{4}\right)\left(1 - \frac{1}{8}\right)(1) = \frac{35}{32} > 1$. A negative cost rewards the expert; it does not violate positivity. The comparator theorem uses cumulative cost $-\frac{1}{2}$ and cumulative magnitude $\frac{3}{2}$ separately. Replacing the magnitude by the signed sum would be invalid.

For $[0, 1]$ losses and known horizon, the uncapped expression $\eta T + \frac{\ln(n)}{\eta}$ suggests $\eta = \sqrt{\frac{\ln(n)}{T}}$. The theorem requires a cap and the case $n = 1$ requires its own branch.

Case	Legal fixed rate	Conclusion
$n = 1$	choose any $0 < \eta \le \frac{1}{2}$, for example $\eta = \frac{1}{2}$	The learner and only expert coincide, so $R_T = 0$ identically. Never substitute $\eta = 0$.
$n \ge 2$	$\eta = \min\left\{\frac{1}{2}, \sqrt{\frac{\ln(n)}{T}}\right\}$	Use the theorem with this one fixed horizon-dependent rate. If the cap is active, combine with the trivial $R_T \le T$ rather than claiming the uncapped algebra.
unknown T	not supplied by this proof	A time-varying schedule or doubling argument needs a separate analysis; $\sqrt{\frac{\ln(n)}{t}}$ is not silently inserted.

IN-CLASS CHECK

Prompt: For $n = 1$, explain why the formula $\sqrt{\frac{\ln(n)}{T}}$ is illegal as a theorem parameter even though the true regret is zero. For $n = 25$ and $T = 100$, compute the capped fixed rate.

Hint: The theorem requires a strictly positive rate; then compare the square-root candidate with $\frac{1}{2}$.

Answer:

For $n = 1$, $\ln n = 0$ makes the square-root candidate 0, outside the domain $0 < \eta \le \frac{1}{2}$. The correct argument is structural: there is only one expert, so learner and comparator have identical loss and $R_T = 0$; one may retain any legal positive rate. For $n = 25$, $T = 100$, $\sqrt{\ln \frac{25}{100}} \approx 0.1794 < \frac{1}{2}$, so the fixed rate is approximately 0.1794.

The proof is now complete, so we can audit formula identity without importing a future online-convex-optimization result. Four familiar-looking operations have different state objects and different guarantees:

Operation	Update shape	What is and is not licensed
course-primary linear MWU	$w_{t+1}(i) = w_t(i)(1 - \eta \ell_t(i))$	This is the update proved above with the fixed-comparator magnitude theorem.
a deferred exponential expert update	$w_{t+1}(i) = w_t(i)e^{-\eta \ell_t(i)}$	Not an alias for the linear factor; its standard name and distinct theorem are confined to the optional appendix.
AdaBoost	$D_{t+1}(j) \propto D_t(j)e^{-\alpha_t y_j h_t(x_j)}$	Updates weights on fixed training examples using signed classification margins; the product-potential pattern transfers, not algorithm identity.

Operation	Update shape	What is and is not licensed
retrieved gradient template	$x_{t+1} = x_t - \eta g_t$ before any feasibility correction	An additive move in a decision space, not linear MWU. Online gradient descent (OGD), its projection, and its regret theorem remain Chapter 9 material.

IN-CLASS CHECK

Prompt: Sort the four formulas by state object and update type. Why is "all are multiplicative weights" false even if all can be analyzed with a potential?

Hint: A proof pattern does not determine an algorithm's state space, feedback, update, or theorem assumptions.

Answer:

> Linear MWU and the deferred exponential expert rule both update expert weights multiplicatively, but with different factors and comparison terms. AdaBoost updates a distribution over fixed training examples using a weak hypothesis and a signed exponential factor. The retrieved gradient template moves a decision vector additively. A potential argument can organize several of these proofs, but it does not make their update equations, domains, or guarantees identical.

The potential theorem gives a reusable result. Choosing among the algorithms, however, still requires matching the theorem's assumptions to the actual prediction problem.

Match assumptions to the guarantee

Each scenario below is deliberately paired with an invalid conclusion. Repair the algorithm and label the kind of guarantee before reading the synthesis.

Scenario	invalid card	repair	guarantee label
binary experts; a perfect expert survives	linear MWU is required		
binary experts; no realizability; deterministic prediction required	additive sublinear regret		
binary experts; private randomization allowed; non-anticipating loss	pathwise regret		
full bounded losses in $[0, 1]$; complete feedback	Halving bound		

Chapter-exit tray: exactly 3	Chapter 9 entry tray: exactly 4

Table 6: An assumption-to-guarantee repair task and two deliberately separate boundary trays.

IN-CLASS CHECK

Prompt: Repair all four scenario rows. Which row needs an expectation label, and which row offers robustness without additive sublinear regret?

Hint: Distinguish realizable mistakes, deterministic comparison, randomized expectation, and bounded-loss regret.

Answer:

> Use Halving for the realizable binary row and label its conclusion a mistake bound. Use deterministic Weighted Majority for the nonrealizable deterministic row and label its result the exact beta-dependent mistake comparison, not additive regret. Use Randomized Weighted Majority for the private-randomization row and label its conclusion mixture loss or expected regret, never pathwise regret. Use linear MWU for bounded full-information losses and label its conclusion a comparator or external-regret bound. The randomized row needs expectation; deterministic Weighted Majority supplies state robustness without the additive rate.

Definition: Selecting an expert-advice algorithm

Check four axes together: whether a perfect expert is guaranteed; whether losses are binary or generally bounded; whether private randomization is permitted; and what current information the adversary observes. Then name the conclusion accurately as a mistake bound, a multiplicative mistake comparison, a mixture or expected-loss guarantee, or an external-regret bound.

IN-CLASS CHECK

Prompt: A client has losses in $[0, 1]$, observes every expert's loss after each commitment, permits private sampling, and supplies no perfect expert. Which core theorem applies, and what additional information condition is needed to interpret an actual sampled action?

Hint: Continuous bounded losses point to the linear update; sampling requires the same current-draw boundary as before.

Answer:

> Apply the linear MWU comparator theorem, or its $[0, 1]$ regret corollary, using a legal fixed η. If an expert is actually sampled rather than charging mixture loss, the current loss vector must be fixed without observing that private sample. The result is then an expected sampled-loss guarantee, not an every-path promise.

Standard language to carry forward

Term or notation	Meaning here	Representative form or condition
Randomized Weighted Majority	Sample an expert from pre-loss normalized weights; analyze mixture or expected sampled loss.	$p_t(i) = \frac{w_t(i)}{\sum_j w_t(j)}$ and a hidden current draw
$p_t \cdot \ell_t$	Registered distribution-weighted round loss.	$\sum_i p_t(i)\ell_t(i)$
linear Multiplicative Weights Update	Multiply each expert weight by a linear function of its bounded current cost.	$w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i))$
weight-potential analysis	Upper-bound total weight through learner loss and lower-bound it through one fixed comparator.	$\Phi_t = \sum_i w_t(i)$
linear MWU comparator bound	Signed-cost theorem before the coarser loss-only regret corollary.	$\sum_t p_t \cdot \ell_t \le \sum_t \ell_t(i) + \eta \sum_t \lvert\ell_t(i)\rvert + \frac{\ln(n)}{\eta}$
fixed-horizon rate	A positive capped rate for $n \ge 2$; $n = 1$ is a separate zero-regret branch.	$\eta = \min\left\{\frac{1}{2}, \sqrt{\frac{\ln(n)}{T}}\right\}$
algorithm selection	Match realizability, loss range, randomization, and adversary information to the guarantee type.	mistake / comparison / mixture or expected / regret

Reconstruct the three Chapter 8 exits

The main results of this chapter are exactly these three items, in order:

No.	Chapter exit	Learner-visible reconstruction
1	external regret	$R_T = \sum_t L_t(\mathcal{A}) - \min_i \sum_t \ell_t(i)$ against one fixed same-sequence comparator
2	linear Multiplicative Weights Update	$p_t(i) = \frac{w_t(i)}{\Phi_t}$ and $w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i))$
3	linear MWU regret theorem	signed comparator-magnitude form and the $[0,1]$ corollary with $0 < \eta \leq \frac{1}{2}$

IN-CLASS CHECK

Prompt: Write the three exits without looking back. Why is the potential technique not a fourth exit?

Hint: The exit list records the approved Chapter boundary, not every useful proof device taught inside it.

Answer:

> The exits are external regret, the linear Multiplicative Weights Update, and the linear MWU regret theorem. Weight-potential analysis is an important internal technique used to establish the theorem, but the approved Chapter boundary exports the theorem and algorithm rather than a fourth list item.

Retrieve the exact Chapter 9 entry

Chapter 9 receives a different four-item list. The first two come from online learning. The last two are prior-course tools retrieved now; they were not introduced by Tutorial 8A and were not added to its eleven-item handoff.

Before using the list, recall the two prior-course meanings. A function f on a convex domain is **convex** when $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$ for $\lambda \in [0,1]$. A **gradient method template** uses a gradient or subgradient to propose a local descent move such as $x - \eta g$. These are retrieved assumptions only. No online convex-optimization algorithm, feasibility projection, regret proof, or lower bound is asserted here.

No.	Chapter 9 may assume	Permitted meaning at entry
1	external regret	Compare cumulative learner loss with one best fixed decision on the same sequence.
2	linear MWU regret theorem	Recognize a finite-simplex comparator theorem with its legal learning-rate domain.
3	convex function	Use the displayed chord inequality as previously learned mathematical knowledge.
4	gradient-method template	Recognize an additive local descent proposal; Chapter 9 must build its online and feasibility machinery.

IN-CLASS CHECK

Prompt: Reconstruct the four items in order. Explain why the Chapter exits and Chapter 9 entry have different cardinalities, and name two results that are still forbidden.

Hint: The next chapter retrieves two old prerequisites and does not yet receive its own future theorems.

Answer:

> The order is external regret; the linear MWU regret theorem; convex function; gradient-method template. The Chapter exits include the linear MWU algorithm, while the next entry replaces that algorithm item with two retrieved optimization prerequisites, so the lists have three and four items respectively. An online gradient descent (OGD) regret theorem and a general square-root lower bound are still forbidden; so are projection and FTRL results not yet taught.

The required core ends here. The optional material below exports nothing to Chapter 9, is unassessed, and consumes zero protected minutes.

Optional self-study appendix

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: What breaks if the adversary sees the learner's current private sampled expert before fixing the current loss?

Answer.

With two constant experts, after seeing the current sample the adversary can assign loss 1 to the sampled expert and 0 to the other. The sampled learner then loses 1 on every round, while one of the two fixed experts loses on at most half the rounds. The conditional equality $E[\ell_t(I_t)|p_t, \ell_t] = p_t \cdot \ell_t$ is no longer licensed because ℓ_t was chosen as a function of I_t. This is the fully adaptive adversary caveat. Knowing the public distribution p_t is allowed in the core model; observing its private realized sample before choosing the loss is the disallowed extra information.

Post-class or self-study. Not a prerequisite for the core path.

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: Is exponential Hedge merely another notation for the linear MWU update proved above?

Answer.

No. Hedge uses $w_{t+1}(i) = w_t(i)e^{-\eta\ell_t(i)}$, whereas the course-primary linear MWU uses $w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i))$. Arora, Hazan, and Kale's Hedge theorem has a distinct second-order term depending on the learner distribution, while the linear theorem above has the fixed comparator's cumulative magnitude. The two algorithms are related multiplicative-update variants, but their factors, domains, and exact guarantees are not interchangeable. This comparison is optional and is not part of the Chapter 9 entry.

Post-class or self-study. Not a prerequisite for the core path.

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: Do portfolios, zero-sum games, boosting, and online optimization all instantiate one identical MWU algorithm?

Answer.

No. An application claim is valid only after identifying its decision set, feedback, loss or payoff, update, and theorem conditions. A portfolio may use a distribution over assets; a zero-sum game may use repeated mixed strategies; AdaBoost updates weights over fixed training examples with a weak-hypothesis-dependent exponential factor; and a gradient method moves a point in a decision set. They can share normalization, regret, or potential ideas without sharing one update equation. The application instances are therefore analogies with domain-specific derivations, not evidence that AdaBoost, linear MWU, and the future Chapter 9 algorithms are identical.

Post-class or self-study. Not a prerequisite for the core path.

Source note

Randomized Weighted Majority, its binary weight identity, Theorem 8.4, and the mixture-loss square-root conclusion follow Mohri, Rostamizadeh, and Talwalkar, **Foundations of Machine Learning**, second edition, Section 8.2.3 (printed pp. 183-186; PDF pp. 200-203). The sampled-loss interpretation uses the explicitly declared course information assumption that the current private sample is not observed before the current loss vector is fixed. The theorem is stated as a mixture or expected-loss guarantee, not as a pathwise claim.

The course-primary linear update, exact total-weight identity, signed-cost comparator theorem, $[0, 1]$ corollary, and optional exponential-update contrast follow Arora, Hazan, and Kale, Sections 2-2.1, especially Figure 1, Theorem 2.1, Corollary 2.2, and Theorem 2.3 (journal pp. 126-129; PDF pp. 6-9). The fixed- horizon cap and separate $n = 1$ branch are made explicit here; no unproved time-varying learning rate is substituted.

The historical Week 8 Learning Sheet fixes the randomized-majority, MWU, and synthesis scope only. Its claims that several domains use one identical update are not retained. AdaBoost's distribution and product proof are recalled from the verified Chapters 6-7 boundary. The convex-function definition and gradient template are retrieved from prior course knowledge solely for the exact Chapter 9 entry; no Chapter 9 online algorithm, projection theorem, regret bound, FTRL claim, or square-root lower bound is used in this Tutorial.